

Brief summary of this function: Plots the different charging phases on graphs in different colours

Detailed explanation of this function: This function was made to separate the different phases on the graphs and plot them in different colours to signify they are not the same phase. Constant Current (CC) is a dark red (so it can be visible against the pink dotted lines (which depict the entire function without the changes in colours), Constant Voltage (CV) is blue, Rest is green, and the Undefined region is cyan. A legend is also added to this function as to show what the different colours mean when they appear on a graph.

```
function plotcycle_modes(time, y, mode) %need to start with function to
establish as helper function

hold on

%using the other helper function (mode) to use to plot the various cycle
%changes
line_1 = plotMode(time, y, mode, "CC", [0.5,0,0]); %made the red darker to
make it more visible against the pink
line_2 = plotMode(time, y, mode, "CV", 'b');
line_3 = plotMode(time, y, mode, "Rest", 'g');
line_4 = plotMode(time, y, mode, "Undefined", 'c'); %cyan

%titling the different lines be reflected in the legend
legend([line_1 line_2 line_3 line_4], 'Constant Current', 'Constant
Voltage', 'Rest', 'Undefined')

grid on

hold off
```